DAY 2: INTRODUCTION TO DYNAMIC ROUTING

Theory + RIP Observation Practical | Cisco Packet Tracer

Theory

What is Dynamic Routing? Dynamic routing is a method in which routers use routing protocols to automatically learn, exchange, and update routes. When the network topology changes, the routing protocol can update routing information without requiring an administrator to manually rewrite every route.

Static Routing vs Dynamic Routing

Point	Static Routing	Dynamic Routing	Real-world analogy
Configuration	Routes are entered manually with ip route.	Routes are learned automatically using protocols such as RIP, OSPF or EIGRP.	Static is like choosing one road manually; dynamic is like Google Maps calculating a route.
Network changes	A normal static route does not automatically find a new path after a failure.	The routing protocol exchanges information and can update routes when topology changes.	If a road closes, Google Maps can calculate another available path.
Scale & overhead	Low routing-update overhead, but manual maintenance grows with the network.	Uses router resources and update traffic, but is easier to maintain in changing networks.	A small Swiggy branch can use fixed paths; a large delivery network benefits from automatic path updates.

Practical: Observe RIP Routing Updates

Use a pre-configured three-router RIP topology. If you need to build one for practice, the following addressing plan can be used:

LAN1 192.168.1.0/24 — R1 === R2 === R3 — LAN3 192.168.3.0/24

R1–R2: 10.0.12.0/30 | R2–R3: 10.0.23.0/30

RIP v2 Configuration (Reference)

R1:

```
enable
configure terminal
router rip
version 2
no auto-summary
network 192.168.1.0
network 10.0.0.0
end
```

R2:

```
enable
configure terminal
router rip
version 2
no auto-summary
network 10.0.0.0
end
```

R3:

```
enable
configure terminal
router rip
version 2
no auto-summary
network 192.168.3.0
network 10.0.0.0
end
```

Task 2: Observe RIP Routes and Metrics

Run the following commands on R1, R2 and R3:

```
show ip route
show ip route rip
show ip protocols
```

In the routing table, routes beginning with R are RIP-learned. Example:

```
R 192.168.3.0/24 [120/2] via 10.0.12.2
```

Here, R = RIP, 120 = RIP administrative distance, and 2 = RIP hop-count metric. RIP uses hop count as its metric; 15 is the maximum usable hop count and 16 means unreachable.

Router	RIP-learned destination	Next hop	Metric (hop count)
R1	_____	_____	_____
R2	_____	_____	_____
R3	_____	_____	_____

Task 3: Simulate a Network Change

Example: shut down R2's interface toward R3. Replace the interface name if your topology uses a different port.

```
R2# configure terminal
R2(config)# interface serial 0/0/1
R2(config-if)# shutdown
R2(config-if)# end
```

Immediately monitor another router repeatedly:

```
R1# show clock
R1# show ip route
R1# show ip route rip

! Repeat show ip route until the unreachable RIP route disappears.
R1# show clock
```

Record your actual Packet Tracer observation rather than inventing a value. RIP normally sends periodic updates about every 30 seconds, while convergence/removal time can depend on RIP timers and how the failure is detected.

Shutdown time	_____
Unreachable route	_____
Route removed at	_____
Observed update/removal time	_____ seconds

Restore the interface:

```
R2# configure terminal
R2(config)# interface serial 0/0/1
R2(config-if)# no shutdown
```

```
R2(config-if)# end
R2# show ip interface brief
```

Task 4: Uber/Zomato Analogy

Dynamic routing is similar to an Uber or Zomato delivery system automatically adjusting a path when conditions change. If a network path becomes unavailable, RIP exchanges routing information and routers update their routing tables. In the same way, a delivery application can recalculate a driver's route when a road becomes unavailable or conditions change. Both approaches reduce the need to manually specify every path change.

Useful Verification Commands

```
show ip interface brief
show ip route
show ip route rip
show ip protocols
show clock
ping <destination-IP>
```